UNIT 1 – INTRODUCTION TO OPERATING SYSTEMS

Syllabus Content:

Introduction to Operating Systems: Definition, Types of Operating system, Real-Time operating system
System Components: System Services, Systems Calls, System Programs, System structure, Virtual Machines, System Design and Implementation, System Generations.

Topic:

- Introduction to Operating System
- Need of Operating System
- Goals and Functions of Operating System

Course Outcomes

- Understand the concept of Operating System
- Explain the need of an Operating System
- Identify the functions of an Operating System
- Understand interaction between hardware and software

Activity
Think for 30 Seconds

- Can a computer work without Operating System?
- Can mobile phones work without Operating System?
- What happens if Operating System is removed?

What is an Operating System?

Definition:
An Operating System (OS) is system software that manages computer hardware and software resources and provides services to application programs.

Keywords

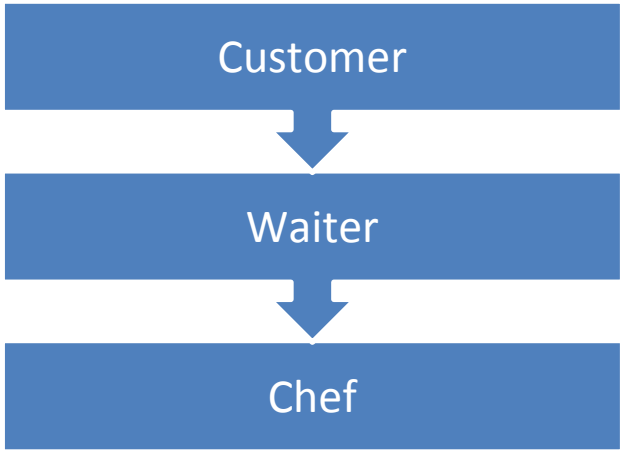
- System Software
- Resource Manager
- Interface
- Services

"Remember"

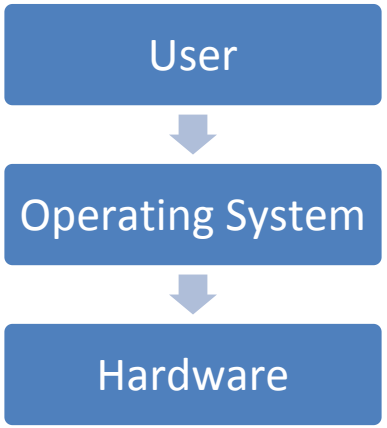
1- The Operating System is a manager.
2- Operating System controls every hardware device inside the computer.

Real Life Analogy

Imagine



Customer never directly enters the kitchen.
Similarly,

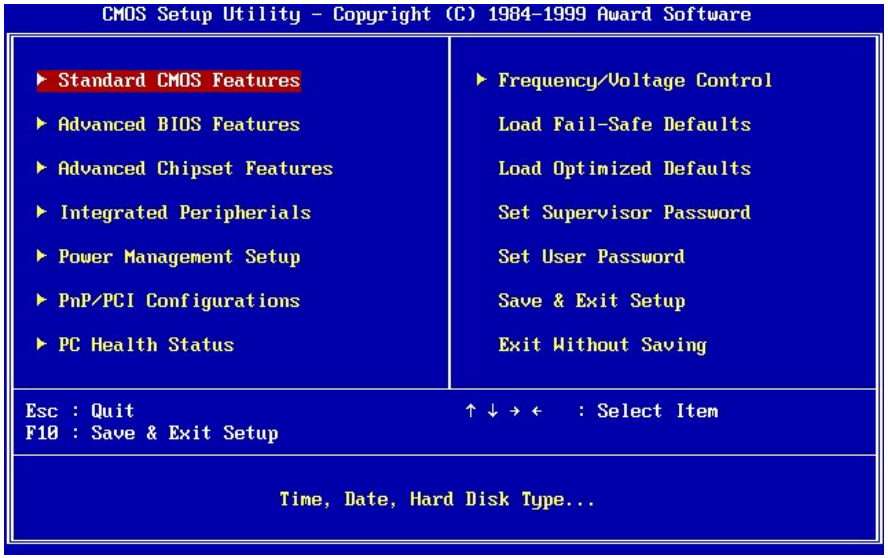


“The user never directly communicates with the processor.
Instead,
the Operating System acts exactly like a waiter in a restaurant.
The waiter takes your order and delivers it to the kitchen.
Similarly,
the OS takes your request and gives it to the CPU.”

Computer Without Operating System

What happens?

- ❌ No login
- ❌ No applications
- ❌ No file management
- ❌ No device drivers
- ❌ No multitasking





Without an Operating System,

1- your computer becomes almost useless.

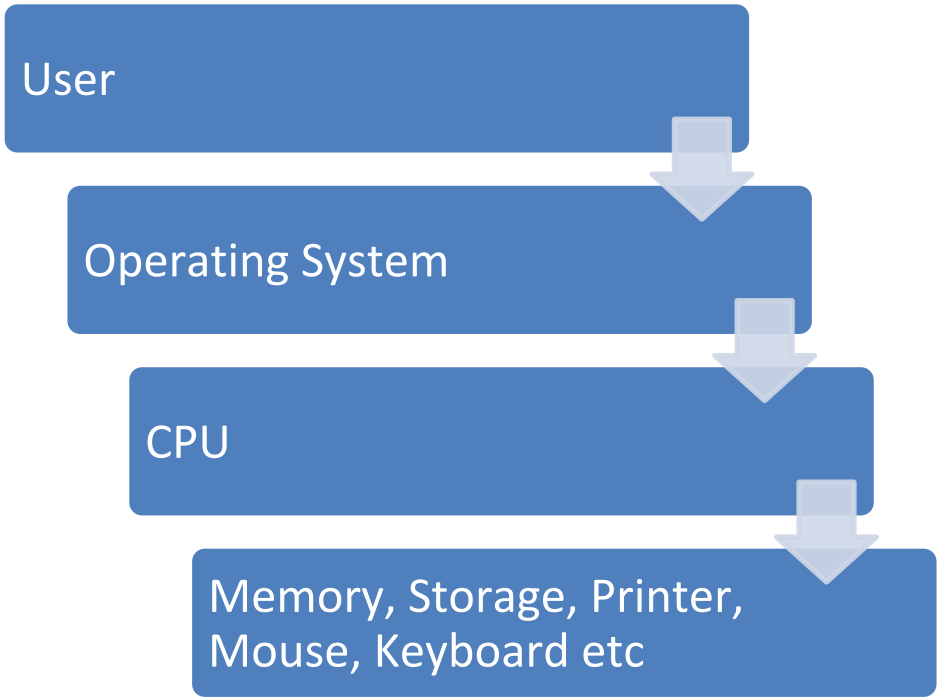
2- You cannot open documents,

3- browse the internet, or even save files.

Computer With Operating System

Operating System provides

- ✔ User Interface
- ✔ Memory Management
- ✔ Process Management
- ✔ Device Management
- ✔ File Management
- ✔ Security



“The OS becomes the bridge between the user and all hardware devices.”

Evolution of Computers

Generation	Example
1st	Vacuum Tubes
2nd	Transistors
3rd	IC
4th	Microprocessor
5th	AI Computers

As hardware became more powerful, Operating Systems also became smarter.

Why Do We Need Operating System?

Because computers have

- CPU
- RAM
- Hard Disk
- Keyboard
- Mouse
- Printer

Someone has to coordinate all of them.
That someone is
Operating System.

Main Functions of Operating System

- Process Management
- Memory Management
- File Management
- Device Management
- Security
- User Interface

Goals of Operating System

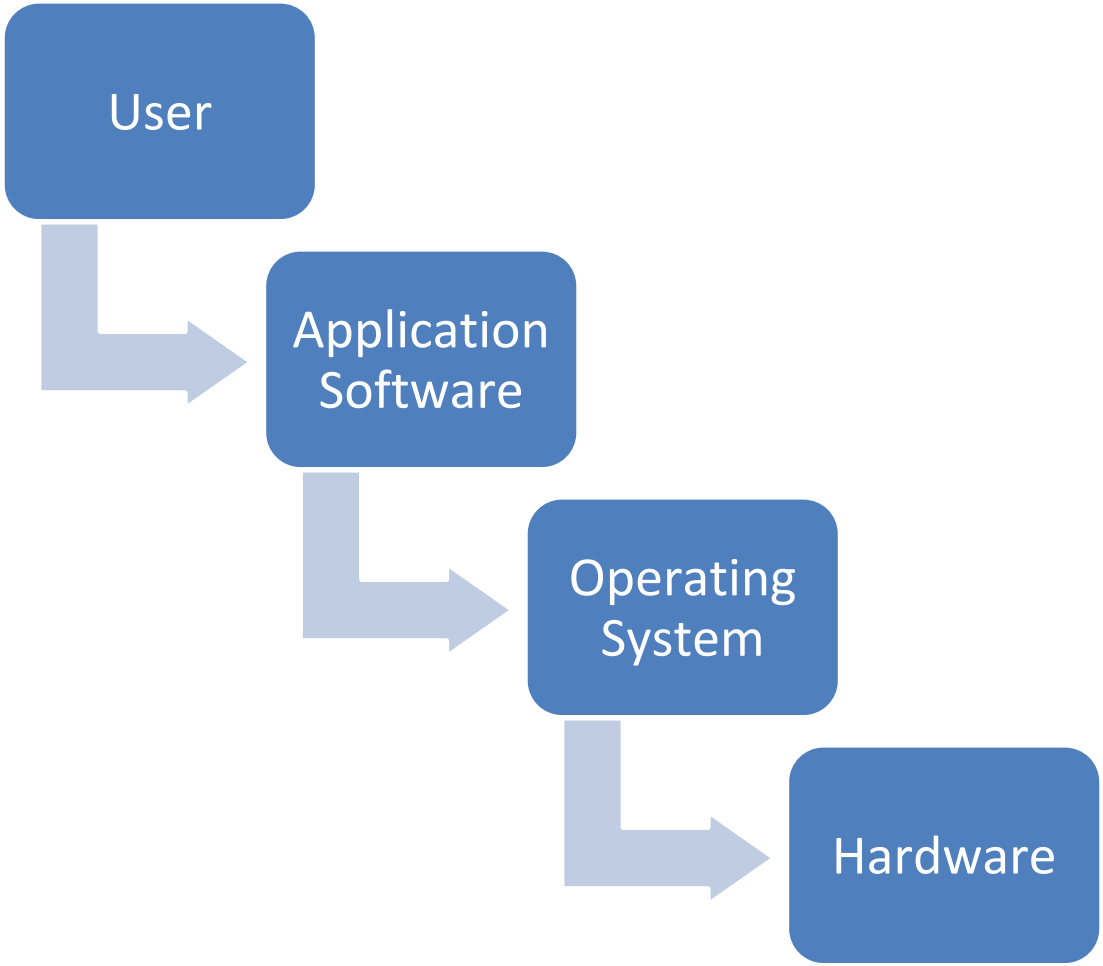
Primary Goals:

Convenience: Make computer easy to use.

Efficiency: Maximum performance.

Ability to Evolve: Easy to upgrade.

Components of Computer System



Types of Users

1- Home User
2- Student
3- Office User
4- Developer
5- Server Administrator

Popular Operating Systems

Windows
Linux
macOS
Android
iOS

Comparison

OS	Developed By	Used In
Windows	Microsoft	PCs
Linux	Community	Servers
macOS	Apple	Macs
Android	Google	Smartphones
iOS	Apple	iPhones

Advantages of Operating System

1. Easy to use
2. Fast execution
3. Security
4. Resource Sharing
5. Error Handling
6. Multitasking

Limitations

1. Expensive
2. Can crash
3. Security attacks
4. Virus
5. Hardware compatibility issues